



VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

FALL SEMESTER 2025-26

BEVD301P-ASIC Design Lab

TASK 4

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Slot: L5+L6

Date:14/10/2025

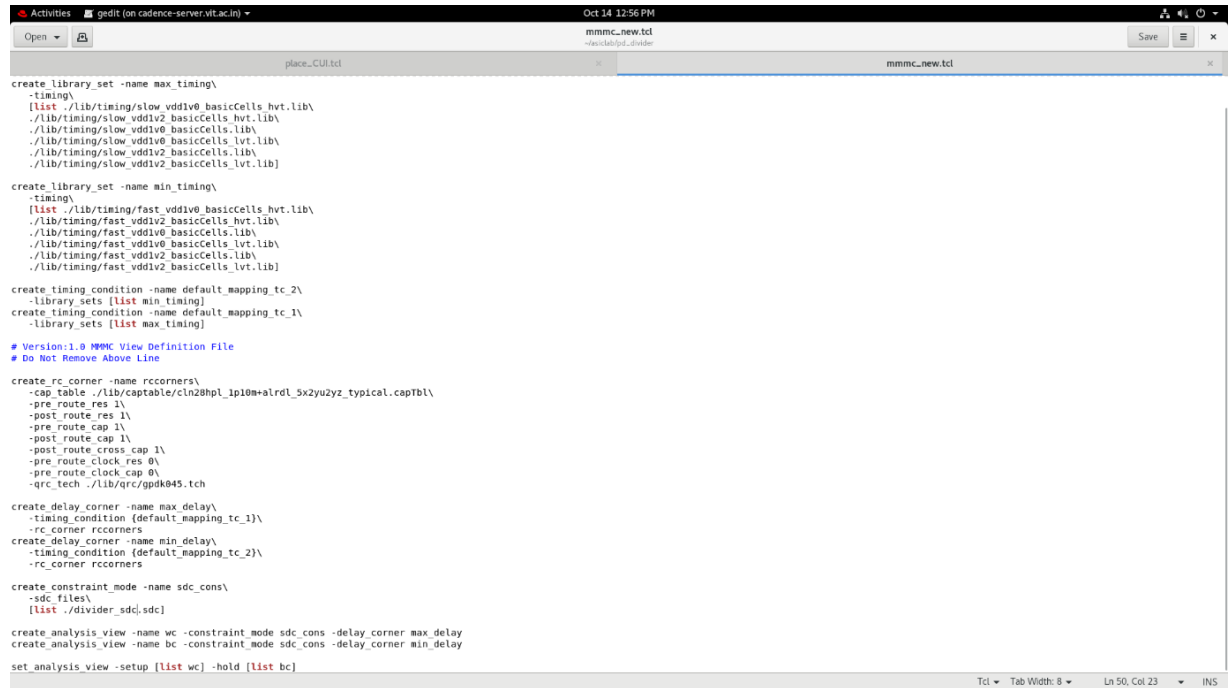
Physical Design-upto Routing

Aim: Performing of floorplanning,placement,cts,routing of my design that is 8-bit divider. And to check with 0 errors and least DRC violations.

Tool Used : Cadence Innovus tool

Outputs:

1) mmmc_new.tcl file



```
create_library_set -name max_timing\
-timing\
[list ./lib/timing/slow_vddlv0 basicCells_hvt.lib\
./lib/timing/slow_vddlv2 basicCells_hvt.lib\
./lib/timing/slow_vddlv0 basicCells.lib\
./lib/timing/slow_vddlv0 basicCells_lvt.lib\
./lib/timing/slow_vddlv2 basicCells.lib\
./lib/timing/slow_vddlv2 basicCells_lvt.lib]

create_library_set -name min_timing\
-timing\
[list ./lib/timing/fast_vddlv0 basicCells_hvt.lib\
./lib/timing/fast_vddlv2 basicCells_hvt.lib\
./lib/timing/fast_vddlv0 basicCells.lib\
./lib/timing/fast_vddlv0 basicCells_lvt.lib\
./lib/timing/fast_vddlv2 basicCells.lib\
./lib/timing/fast_vddlv2 basicCells_lvt.lib]

create_timing_condition -name default_mapping_tc_2\
-library_sets [list min_timing]
create_timing_condition -name default_mapping_tc_1\
-library_sets [list max_timing]

# Version:1.0 MPMC View Definition File
# Do Not Remove Above Line

create_rc_corner -name rccorners\
-cap_table ./lib/captable/cin28hpl_1p10m+alrdl_5x2y2y2_typical.capTbl\
-pre_route_res 1\
-post_route_res 1\
-pre_route_cap 1\
-post_route_cap 1\
-post_route_cross_cap 1\
-pre_route_clock_res 0\
-pre_route_clock_cap 0\
-qrc_tech ./lib/qrc/gpdk045.tch

create_delay_corner -name max_delay\
-timing_condition (default_mapping_tc_1)\
-rc_corner rccorners
create_delay_corner -name min_delay\
-timing_condition (default_mapping_tc_2)\
-rc_corner rccorners

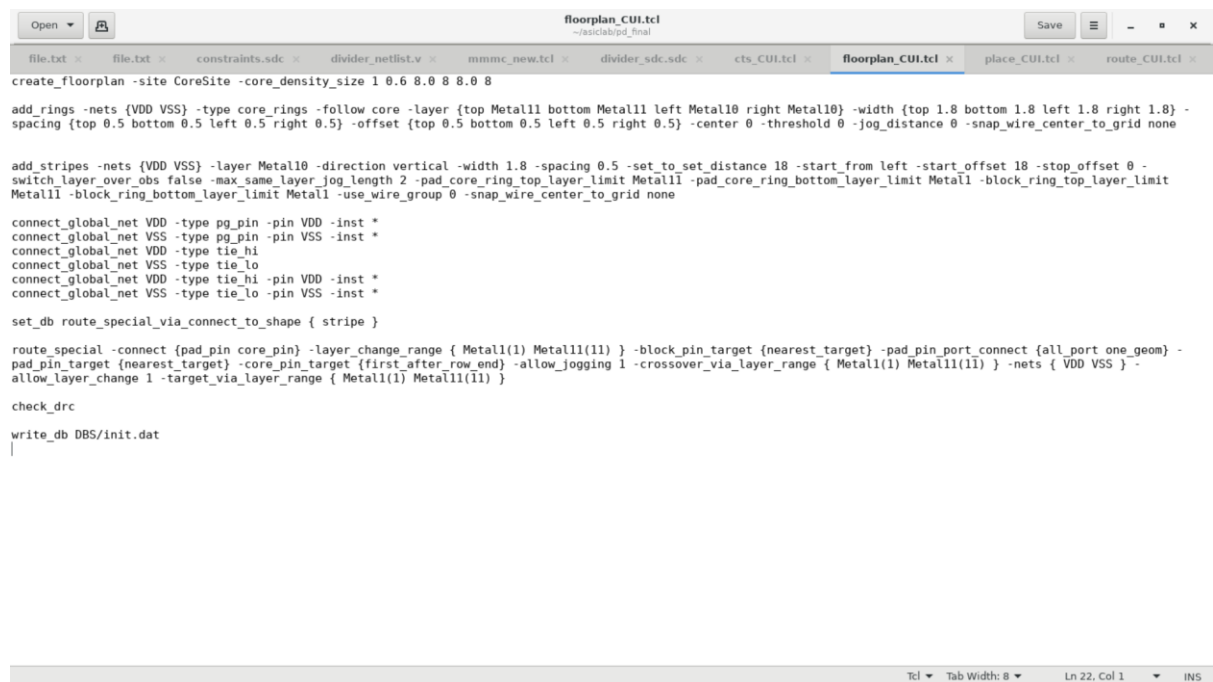
create_constraint_mode -name sdc_cons\
-sdc_files\
[list ./divider_sdc.sdc]

create_analysis_view -name wc -constraint_mode sdc_cons -delay_corner max_delay
create_analysis_view -name bc -constraint_mode sdc_cons -delay_corner min_delay

set_analysis_view -setup [list wc] -hold [list bc]
```

2) floor planning

i) script



```
create_floorplan -site CoreSite -core_density_size 1 0.6 8.0 8.0 8.0 8

add_rings -nets {VDD VSS} -type core_rings -follow core -layer {top Metall1 bottom Metall1 left Metall0 right Metall0} -width {top 1.8 bottom 1.8 left 1.8 right 1.8} -spacing {top 0.5 bottom 0.5 left 0.5 right 0.5} -offset {top 0.5 bottom 0.5 left 0.5 right 0.5} -center 0 -threshold 0 -jog_distance 0 -snap_wire_center_to_grid none

add_stripes -nets {VDD VSS} -layer Metall0 -direction vertical -width 1.8 -spacing 0.5 -set_to_set_distance 18 -start_from left -start_offset 18 -stop_offset 0 -switch_layer_over_obs false -max_same_layer_jog_length 2 -pad_core_ring_top_layer_limit Metall1 -pad_core_ring_bottom_layer_limit Metall1 -block_ring_bottom_layer_limit Metall1 -block_ring_top_layer_limit Metall1 -block_ring_bottom_layer_limit Metall1 -use_wire_group 0 -snap_wire_center_to_grid none

connect_global_net VDD -type pg_pin -pin VDD -inst *
connect_global_net VSS -type pg_pin -pin VSS -inst *
connect_global_net VDD -type tie_hi
connect_global_net VSS -type tie_lo
connect_global_net VDD -type tie_hi -pin VDD -inst *
connect_global_net VSS -type tie_lo -pin VSS -inst *

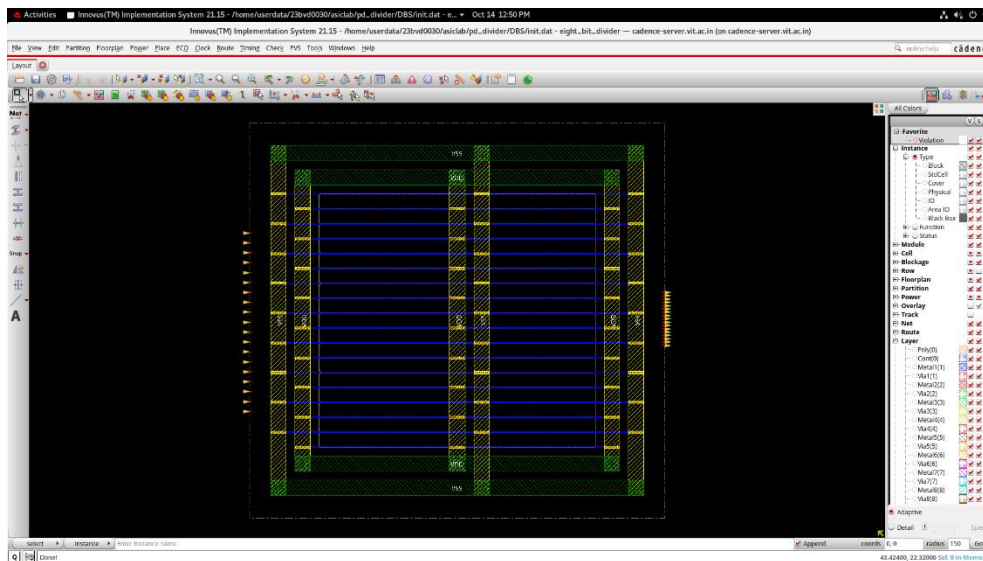
set_db route_special_via_connect_to_shape { stripe }

route_special -connect {pad_pin core_pin} -layer_change_range { Metall1 Metall11 } -block_pin_target {nearest_target} -pad_pin_port_connect {all_port one_geom} -pad_pin_target {nearest_target} -core_pin_target {first_after_row_end} -allow_jogging 1 -crossover_via_layer_range { Metall1 Metall11 } -nets { VDD VSS } -allow_layer_change 1 -target_via_layer_range { Metall1 Metall11 }

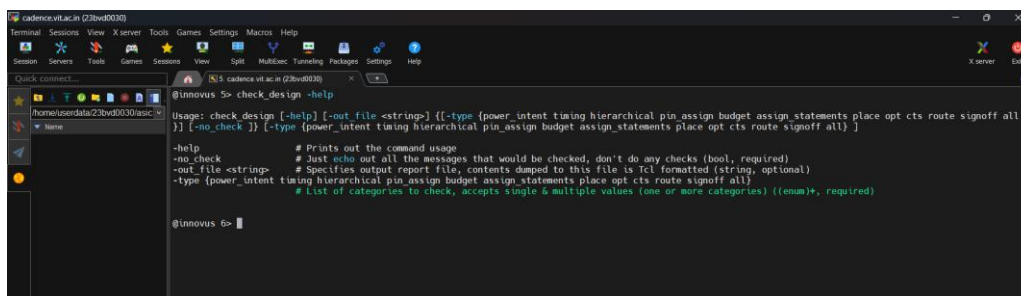
check_drc

write_db DBS/init.dat
```

ii) picture of the floorplan

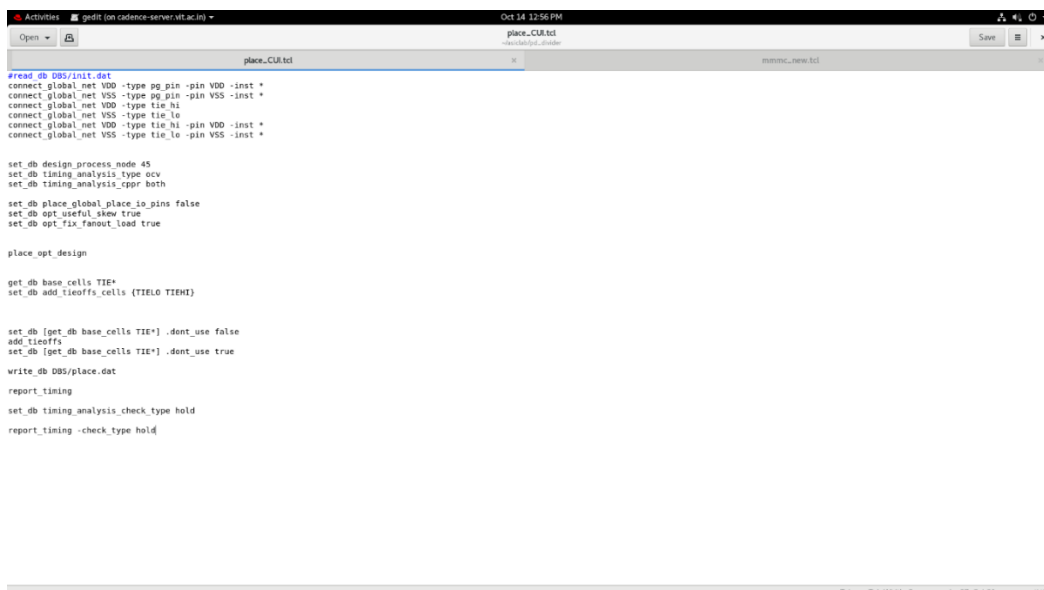


iii) check_design -help



3) placement

i) script



ii)time_design -pre_cts

a)setup

```
cadence.vt.ac.in (23bv0030)
End delay calculation (fullDC). (MEM=2041.24 CPU=0:00:00.2 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:00.3 real=0:00:00.0 totSessionCpu=0:00:29.8 mem=2041.2M)

-----
time_design Summary
-----

Setup views included:
wc

-----
Setup mode | all | reg2reg | reg2gate | default |
-----
WNS (ns): | -0.206 | -0.206 | 0.020 | 0.015 |
TNS (ns): | -1.573 | -1.573 | 0.000 | 0.000 |
Violating Paths: | 9 | 0 | 0 | 0 |
All Paths: | 115 | 39 | 4 | 107 |
-----

-----
DRVs | Real | Total |
-----
Nr nets(terms) | Worst Vio | Nr nets(terms) |
-----
max_cap | 0 (0) | 0.000 | 0 (0) |
max_tran | 0 (0) | 0.000 | 0 (0) |
max_fanout | 0 (0) | 0 | 0 (0) |
max_length | 0 (0) | 0 | 0 (0) |
-----

Density: 47.32%
Routing Overflow: 0.00% H and 0.00% V
Reported timing to dir ./timingReports
Total CPU time: 0.30 sec
Total Real time: 4.0 sec
Total Memory Usage: 2028.05146 Mbytes
*** time_design #1 [finish] : cpu/real = 0:00:01.0/0:00:03.4 (0.3), totSession cpu/real = 0:00:30.0/0:04:04.4 (0.1), mem = 2028.7M
@innovus 7>
```

b)hold

```
cadence.vt.ac.in (23bv0030)
# Design Mode: 45nm
# Analysis Mode: HMC DCV
# Parasitics Mode: No SPDF/RCDB
# Signoff Settings: ST Off
*****
Start delay calculation (fullDC) (1.T). (MEM=2005.01)
*** Calculating scaling factor for min timing libraries using the default operating condition of each library.
Total number of fetched objects: 327
ANS (INFO): Total number of nets for which stage creation was skipped for all views: 0
End delay calculation. (MEM=2061.7 CPU=0:00:00.0 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=2061.7 CPU=0:00:00.1 REAL=0:00:00.0)
Turning on Fast DC mode.
*** Done Building Timing Graph (cpu=0:00:00.2 real=0:00:00.0 totSessionCpu=0:00:32.3 mem=2069.7M)

-----
time_design Summary
-----

Hold views included:
bc

-----
Hold mode | all | reg2reg | reg2gate | default |
-----
WNS (ns): | -0.241 | -0.241 | -0.224 | -0.210 |
TNS (ns): | -21.588 | -0.205 | -0.272 | -17.840 |
Violating Paths: | 115 | 39 | 4 | 107 |
All Paths: | 115 | 39 | 4 | 107 |
-----

Density: 47.32%
Routing Overflow: 0.00% H and 0.00% V
Reported timing to dir ./timingReports
Total CPU time: 0.58 sec
Total Real time: 1.0 sec
Total Memory Usage: 1986.0 Mbytes
*** time_design #2 [finish] : cpu/real = 0:00:00.6/0:00:00.6 (1.0), totSession cpu/real = 0:00:32.5/0:05:19.7 (0.1), mem = 1986.0M
@innovus 8>
```

iii)check_place

```
cadence.vt.ac.in (23bv0030)
@innovus 3> check_place
Begin checking placement ... (start mem=1808.0M, init mem=1808.0M)
*info: Placed = 394
*info: Unplaced = 0
Placement Density: 47.33%(812/1715)
Placement Density (including fixed std cells): 47.33%(812/1715)
Finished check_place (total: cpu=0:00:00.3, real=0:00:00.0; vio checks: cpu=0:00:00.0, real=0:00:00.0; mem=1808.0M)
@innovus 4>
```

4)cts

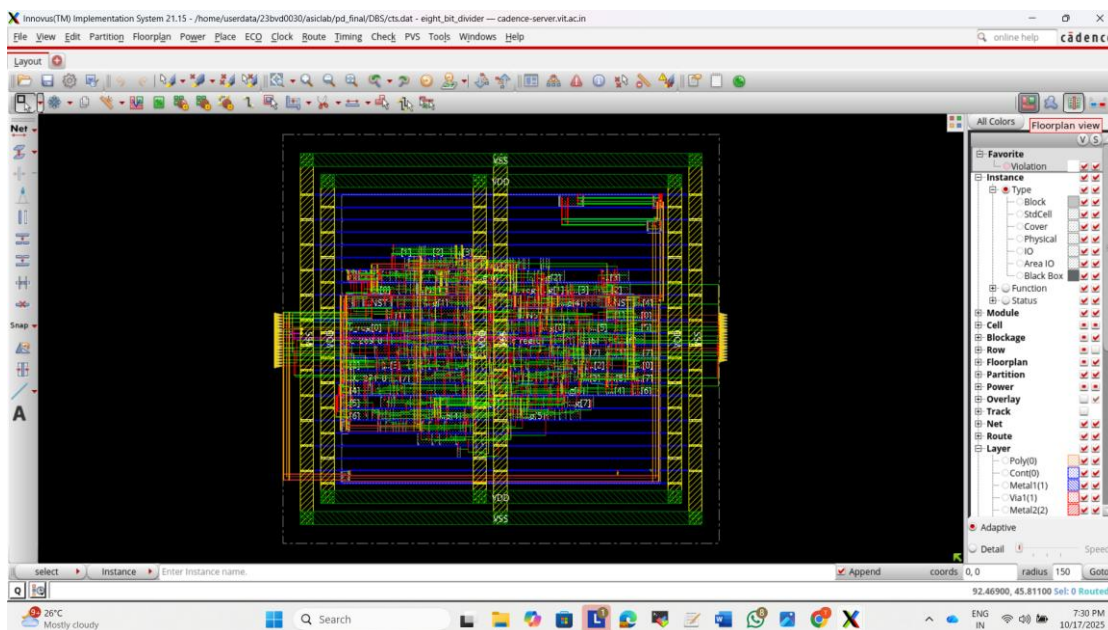
i)script

```
Activities cts_CUI.tcl (~/.asichab/pd_divider) - gedit (on cadence-server.vit.ac.in) Oct 14 1:30 PM
Open cts_CUI.tcl ~/.asichab/pd_divider Save
place_CUI.tcl mmmc_new.tcl cts_CUI.tcl floorplan_CUI.tcl init_CUI.tcl route_CUI.tcl

#read db DBS/place.dat
extract_rc
write_parasitics -spcf_file divider.spcf -rc_corner rccorners
set_db route_design_detail_use_multi_cut_via_effort high
create_route_rule -width {Metal1 0.12 Metal2 0.14 Metal3 0.14 Metal4 0.14 Metal5 0.14 Metal6 0.14 Metal7 0.14 Metal8 0.14 Metal9 0.14 } \
-spacing {Metal1 0.12 Metal2 0.14 Metal3 0.14 Metal4 0.14 Metal5 0.14 Metal6 0.14 Metal7 0.14 Metal8 0.14 Metal9 0.14 } -name 2w2s
create_route_type -name clkroute -route_rule 2w2s -bottom_preferred_layer Metal5 -top_preferred_layer Metal6 -shield_net VSS
set_db cts_route_type_top clkroute
set_db cts_route_type_trunk clkroute
set_db cts_route_type_leaf clkroute
get_db route_design_with_litho_driven
set_db opt_useful_skew_ccopt_medium
set_db cts_buffer_cells "CLKBUF12 CLKBUF16 CLKBUF18 CLKBUF19"
set_db cts_inverter_cells "CLKINV12 CLKINV16 CLKINV18 CLKINV19"
set_db cts_use_inverters true
set_db cts_clock_gating_cells TLATNTSCA*
create_clock_tree_spec -out_file ccopt_cui.spec
source ccopt_cui.spec
ccopt_design
write_db DBS/cts.dat
time_design -post_cts
opt_design -post_cts
report_clock_trees -out_file ../rclk_full.rpt
report_skew_groups -out_file ../rskg_full.rpt
report_timing -late
report_timing -early
set_db opt_fix_hold_allow_setup_tns_degradation false
set_db opt_fix_hold_ignore_path_groups default
opt_design -post_cts -hold -report_dir RPT -report_prefix postcts_hold
write_db DBS/postcts_hold.dat

Tcl Tab Width: 8 Ln 50, Col 1 INS
```

ii)picture of the design



iii)before optimizing design

a)hold

time_design -post_cts -hold

```
# Design Name: eight_bit_divider
# Design Mode: 45nm
# Analysis Mode: PVT: DCV
# Parasitics Mode: No SPT/RCDB
# Signoff Settings: SI Off
*****
Start delay calculation (fullDC) (1 T). (MEM=2737.63)
*** Calculating scaling factor for min timing libraries using the default operating condition of each library.
Total number of fetched objects 337
MAE_INF0: total number of nets for which stage creation was skipped for all views 0
End delay calculation. (MEM=2789.79 CPU=0:00:00.0 REAL=0:00:00.0)
End delay calculation (fullDC): (MEM=2789.79 CPU=0:00:00.1 REAL=0:00:00.0)
*** Done Building Timing Graph (cpus=0:00:00.2 real=0:00:01.0 totSessionCpus=0:02:17 mem=2013.8M)

-----
time_design Summary
-----
Hold views included:
bc:

+-----+-----+-----+-----+-----+
| Hold mode | all | reg2reg | reg2cgate | default |
+-----+-----+-----+-----+-----+
| WNS (ns): | -0.181 | -0.061 | 0.027 | -0.101 |
| TNS (ns): | -5.972 | -0.097 | 0.888 | -5.469 |
| Violating Paths: | 113 | 28 | 0 | 104 |
| All Paths: | 115 | 39 | 4 | 107 |
+-----+-----+-----+-----+-----+

Density: 48.444%
Routing Overflow: 0.00% H and 0.00% V
Reported timing to dir ./timingReports
Total CPU time: 0.59 sec
Total Real time: 1.0 sec
Total Memory Usage: 2744.824219 Mbytes
*** time_design #4 [finish]: cpu/real = 0:00:00.6/0:00:00.6 (1.0), totSession cpu/real = 0:02:17.3/0:13:08.1 (0.2), mem = 2744.8M
@innovus 60>
```

b)setup

time_design -post_cts -setup

```
End delay calculation (fullDC): (MEM=2787.09 CPU=0:00:00.1 REAL=0:00:00.0)
*** Done Building Timing Graph (cpus=0:00:00.2 real=0:00:00.0 totSessionCpus=0:02:16 mem=2011.9M)

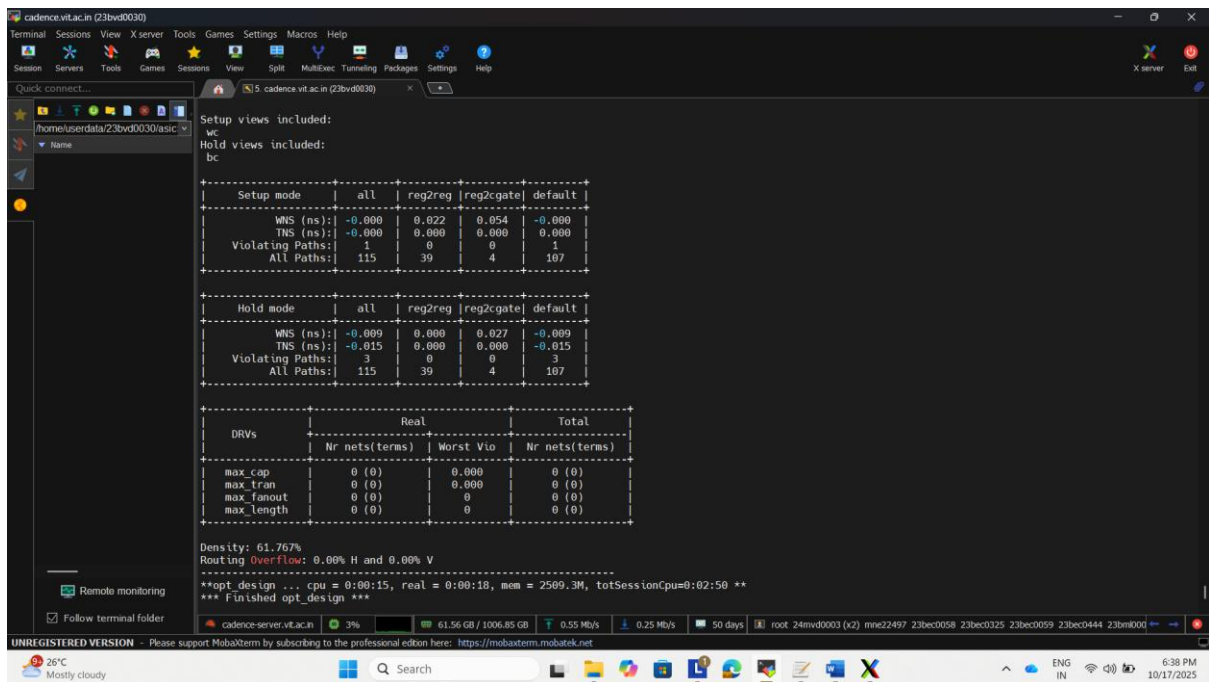
-----
time_design Summary
-----
Setup views included:
uc:

+-----+-----+-----+-----+-----+
| Setup mode | all | reg2reg | reg2cgate | default |
+-----+-----+-----+-----+-----+
| WNS (ns): | 0.007 | 0.007 | 0.044 | 0.030 |
| TNS (ns): | 0.000 | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 | 0 |
| All Paths: | 115 | 39 | 4 | 107 |
+-----+-----+-----+-----+-----+

+-----+-----+-----+-----+-----+
| DRVs | Real | Worst Vio | total |
|-----|-----|-----|-----|
| max cap | 0 (0) | 0.000 | 0 (0) |
| max tran | 0 (0) | 0.000 | 0 (0) |
| max fanout | 0 (0) | 0 | 0 (0) |
| max length | 0 | 0 | 0 (0) |
+-----+-----+-----+-----+-----+

Density: 48.444%
Routing Overflow: 0.00% H and 0.00% V
Reported timing to dir ./timingReports
Total CPU time: 0.75 sec
Total Real time: 4.8 sec
Total Memory Usage: 2743.61198 Mbytes
*** time_design #3 [finish]: cpu/real = 0:00:00.7/0:00:03.2 (8.2), totSession cpu/real = 0:02:16.3/0:12:40.9 (0.2), mem = 2743.6M
@innovus 50>
```

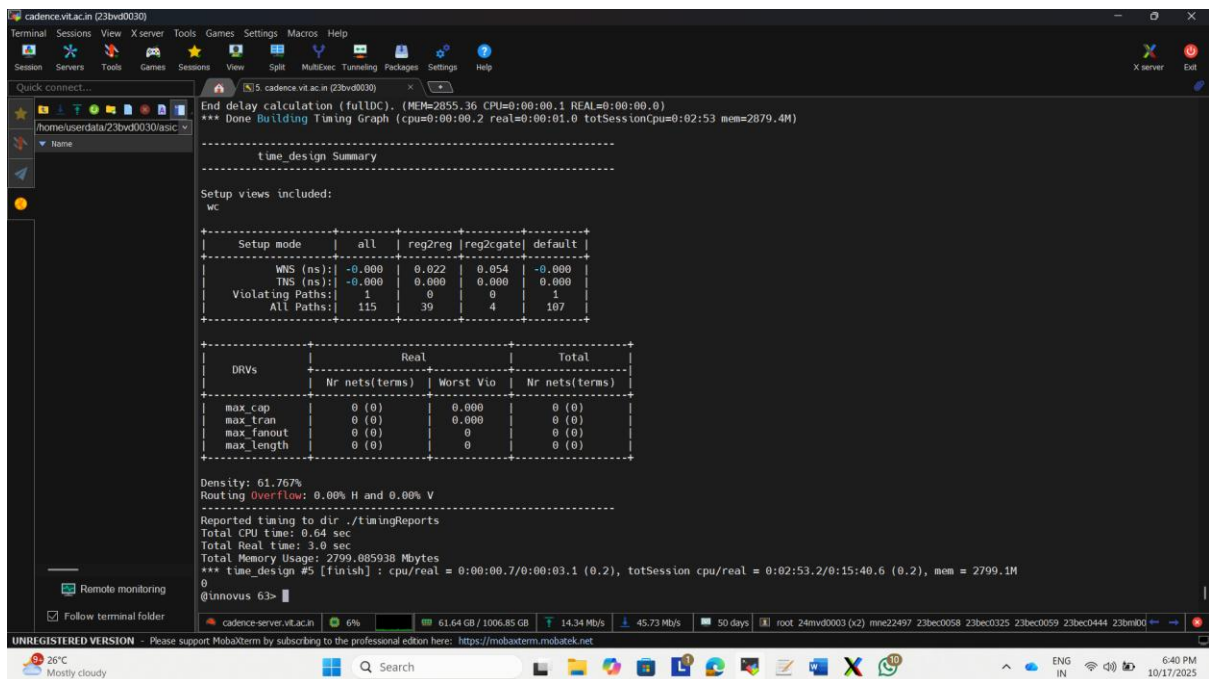
iv)opt_design -post_cts



```
cadence.vt.ac.in (23bvd0030)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
home/userdata/23bvd0030/asic
Name
Setup views included:
wc
Hold views included:
bc
-----
Setup mode | all | reg2reg | reg2cgate | default |
-----
WNS (ns): | -0.000 | 0.022 | 0.054 | -0.000 |
TNS (ns): | -0.000 | 0.000 | 0.000 | 0.000 |
Violating Paths: | 1 | 0 | 0 | 1 |
All Paths: | 115 | 39 | 4 | 107 |
-----
Hold mode | all | reg2reg | reg2cgate | default |
-----
WNS (ns): | -0.009 | 0.000 | 0.027 | -0.009 |
TNS (ns): | -0.015 | 0.000 | 0.000 | -0.015 |
Violating Paths: | 3 | 0 | 0 | 3 |
All Paths: | 115 | 39 | 4 | 107 |
-----
DRVs | Real | Total |
-----
Nr nets (terms) | Worst Vio | Nr nets (terms) |
-----
max_cap | 0 (0) | 0.000 | 0 (0) |
max_tran | 0 (0) | 0.000 | 0 (0) |
max_fanout | 0 (0) | 0 | 0 (0) |
max_length | 0 (0) | 0 | 0 (0) |
-----
Density: 61.767%
Routing Overflow: 0.00% H and 0.00% V
-----
**opt_design ... cpu = 0:00:15, real = 0:00:18, mem = 2509.3M, totSessionCpu=0:02:50 **
*** Finished opt_design ***
UNREGISTERED VERSION - Please support Mobaxterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net
26°C Mostly cloudy
Search
6:38 PM 10/17/2025
```

v)after opt design

a)setup



```
cadence.vt.ac.in (23bvd0030)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
home/userdata/23bvd0030/asic
Name
End delay calculation (fullDC). (MEM=2855.36 CPU=0:00:00.1 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:00.2 real=0:00:01.0 totSessionCpu=0:02:53 mem=2879.4M)
-----
time_design Summary
-----
Setup views included:
wc
-----
Setup mode | all | reg2reg | reg2cgate | default |
-----
WNS (ns): | -0.000 | 0.022 | 0.054 | -0.000 |
TNS (ns): | -0.000 | 0.000 | 0.000 | 0.000 |
Violating Paths: | 1 | 0 | 0 | 1 |
All Paths: | 115 | 39 | 4 | 107 |
-----
DRVs | Real | Total |
-----
Nr nets (terms) | Worst Vio | Nr nets (terms) |
-----
max_cap | 0 (0) | 0.000 | 0 (0) |
max_tran | 0 (0) | 0.000 | 0 (0) |
max_fanout | 0 (0) | 0 | 0 (0) |
max_length | 0 (0) | 0 | 0 (0) |
-----
Density: 61.767%
Routing Overflow: 0.00% H and 0.00% V
-----
Reported timing to dir ./timingReports
Total CPU time: 0.64 sec
Total Real time: 3.0 sec
Total Memory Usage: 2799.085938 Mbytes
*** time_design #5 [finish] : cpu/real = 0:00:00.7/0:00:03.1 (0.2), totSession cpu/real = 0:02:53.2/0:15:40.6 (0.2), mem = 2799.1M
0
@Unnovus 63>
UNREGISTERED VERSION - Please support Mobaxterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net
26°C Mostly cloudy
Search
6:40 PM 10/17/2025
```

b)hold

```
# Design Name: eight_bit_divider
# Design Mode: 45nm
# Analysis Mode: HMC OCV
# Parasitics Mode: No SPEF/RCDB
# Signoff Settings: SI Off
#####
*** calculating scaling factor for min_timing libraries using the default operating condition of each library.
Total number of fetched objects 478
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
Start delay calculation (failDC) (1 T). (MEM=2883.1)
End delay calculation. (MEM=2855.26 CPU=0:00:00.0 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:00.2 real=0:00:00.0 totSessionCpu=0:02:54 mem=2879.3M)

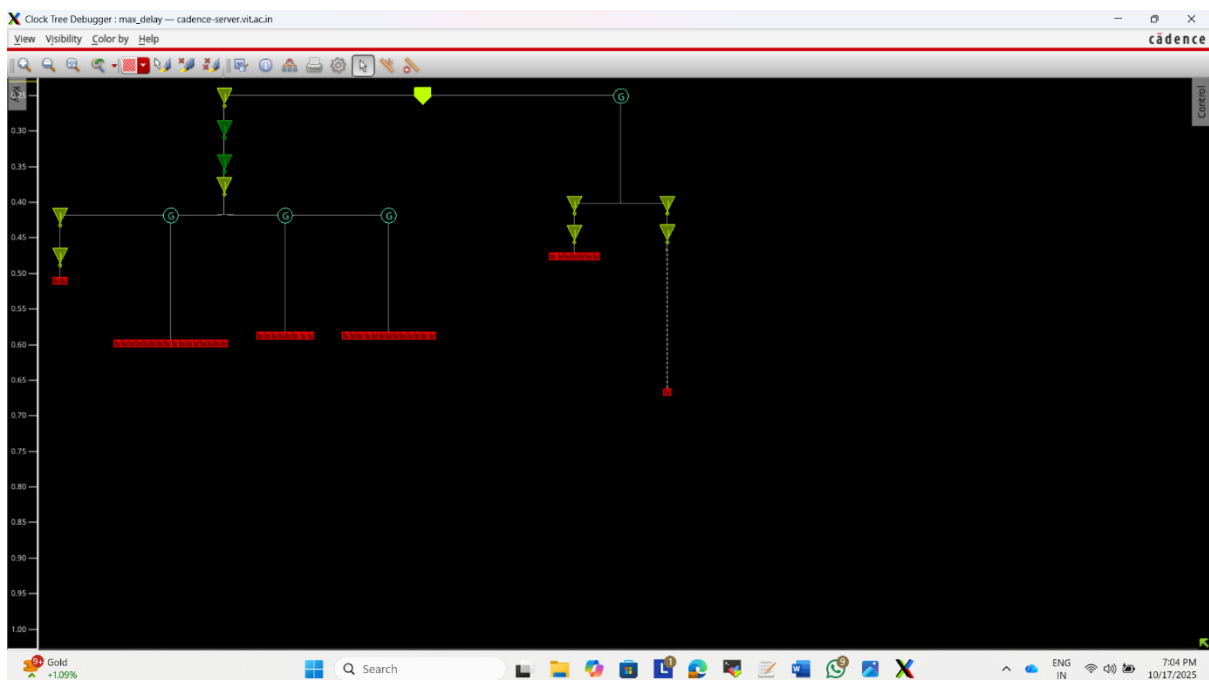
-----
time_design Summary
-----

Hold views included:
bc

-----
+-----+-----+-----+-----+-----+
| Hold mode | all | reg2reg | reg2cgate | default |
+-----+-----+-----+-----+-----+
| WNS (ns): | -0.009 | 0.000 | 0.027 | -0.009 |
| THS (ns): | -0.015 | 0.000 | 0.000 | -0.015 |
| Violating Paths: | 3 | 0 | 0 | 3 |
| All Paths: | 115 | 39 | 4 | 107 |
+-----+-----+-----+-----+-----+

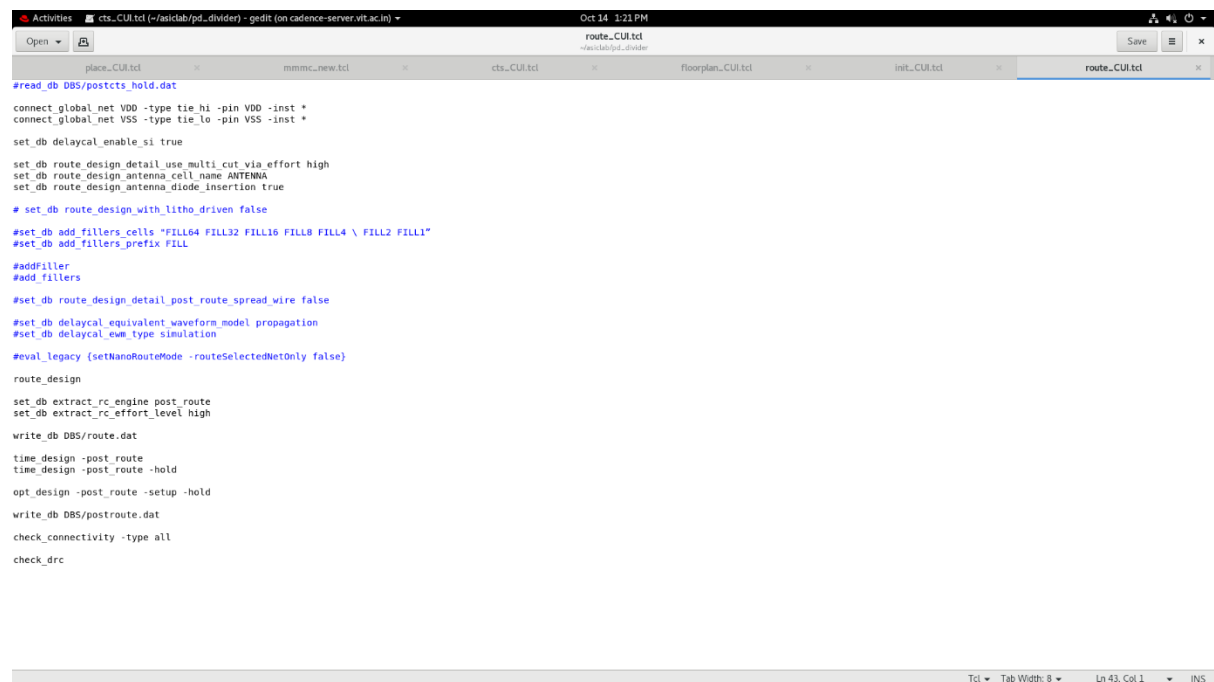
Density: 61.767%
Routing Overflow: 0.00% H and 0.00% V
Reported timing to dir ./timingReports
Total CPU time: 0:50 sec
Total Real time: 0:0 sec
Total Memory Usage: 2888.292969 Mbytes
*** time_design #6 [finish] : cpu/real = 0:00:00.6/0:00:00.6 (1.0), totSession cpu/real = 0:02:54.0/0:15:52.7 (0.2), mem = 2888.3M
@unnovus 64> █
```

vi)clock tree synthesis



5)route

i)script



```
Oct 14 1:21 PM
route_CUI.tcl
#read_db DBS/postcts_hold.dat

connect_global_net VDD -type tie hi -pin VDD -inst *
connect_global_net VSS -type tie lo -pin VSS -inst *

set_db delaycal_enable_si true

set_db route_design_detail use_multi_cut_via_effort_high
set_db route_design_antenna_cell_name ANTENNA
set_db route_design_antenna_diode_insertion true

# set_db route_design_with_litho_driven false

#set_db add_fillers_cells "FILL64 FILL32 FILL16 FILL8 FILL4 \ FILL2 FILL1"
#set_db add_fillers_prefix FILL

#addFiller
#add_fillers

#set_db route_design_detail_post_route_spread_wire false

#set_db delaycal_equivalent_waveform_model_propagation
#set_db delaycal_ewm_type simulation

#eval_legacy {setNanoRouteMode -routeSelectedNetOnly false}

route_design

set_db extract_rc_engine post_route
set_db extract_rc_effort_level high

write_db DBS/route.dat

time_design -post_route
time_design -post_route -hold

opt_design -post_route -setup -hold

write_db DBS/postroute.dat

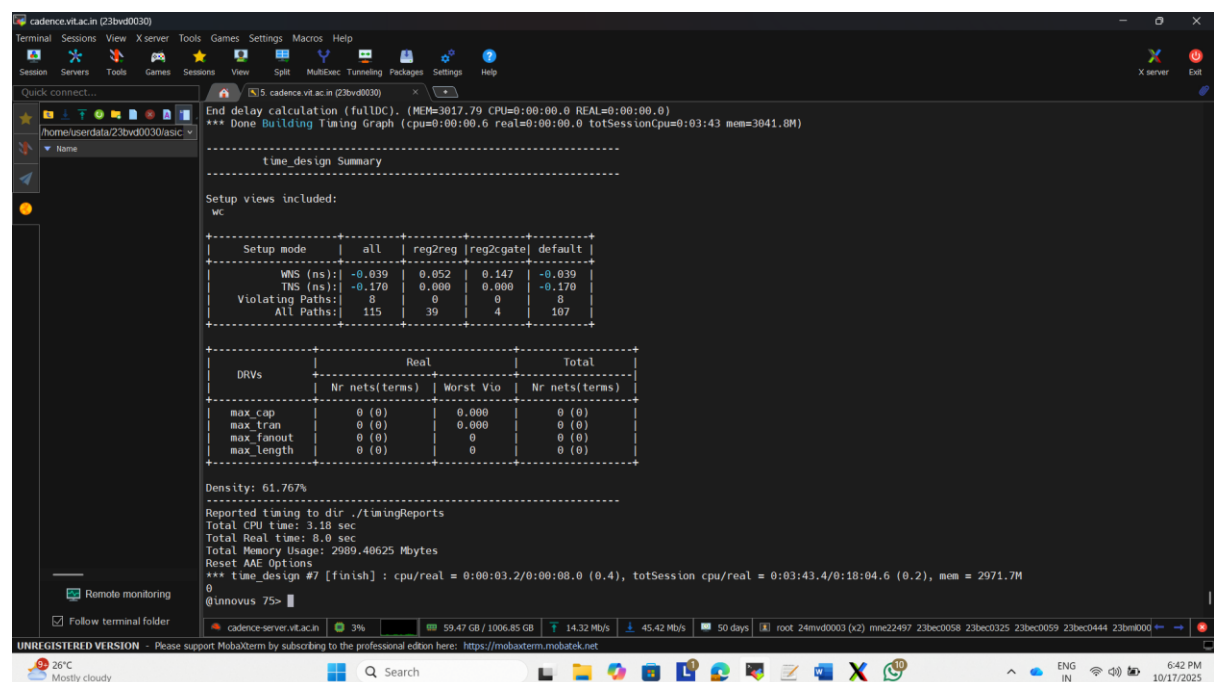
check_connectivity -type all

check_drc
```

ii) time design before optimization

a)setup

time_design -post_route -setup



```
cadence.vita.in (23bvd0030)
*** Done Building Timing Graph (cpu=0:00:00.0 real=0:00:00.0 totSessionCpu=0:03:43 mem=3841.8M)

-----
time_design Summary
-----

Setup views included:
wc

-----
Setup mode | all | reg2reg | reg2cgate | default |
-----
WNS (ns): | -0.039 | 0.052 | 0.147 | -0.039 |
TNS (ns): | -0.170 | 0.000 | 0.000 | -0.170 |
Violating Paths: | 8 | 0 | 0 | 8 |
All Paths: | 115 | 39 | 4 | 107 |
-----

-----
DRVs | Real | Total |
-----
Nr nets(terms) | Worst Vio | Nr nets(terms) |
-----
max_cap | 0 (0) | 0.000 | 0 (0) |
max_tran | 0 (0) | 0.000 | 0 (0) |
max_fanout | 0 (0) | 0 | 0 (0) |
max_length | 0 (0) | 0 | 0 (0) |
-----

Density: 61.767%

Reported timing to dir ./timingReports
Total CPU time: 3.18 sec
Total Real time: 8.0 sec
Total Memory Usage: 2989.48625 Mbytes
Reset AIE Options
*** time_design #7 [finish] : cpu/real = 0:00:03.2/0:00:08.0 (0.4), totSession cpu/real = 0:03:43.4/0:18:04.6 (0.2), mem = 2971.7M
@innovus 75> █
```

b)hold

time_design -post_route -hold

```
cadence.vt.ac.in (23bvd0030)
Loading CTE timing window with TnfLowType 0... (CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 3037.5M)
Add other clocks and setupClocksAEClockMapping during iter 1
Loading CTE timing window is completed (CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 3037.5M)
Starting SI iteration 2
Start delay calculation (fullDC) (1 T). (MEM=2977.58)
Glitch Analysis: View bc -- Total Number of Nets Skipped = 2.
Glitch Analysis: View bc -- Total Number of Nets Analyzed = 478.
Total number of fetched objects 478
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 481, 34.5 percent of the nets selected for SI analysis
End delay calculation. (MEM=3021.27 CPU=0:00:00.0 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=3021.27 CPU=0:00:00.1 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:00.5 real=0:00:00.0 totSessionCpu=0:03:44 mem=3045.3M)

-----
time_design Summary
-----

Hold views included:
bc

+-----+-----+-----+-----+
| Hold mode | all | reg2reg | reg2cgate | default |
+-----+-----+-----+-----+
| WNS (ns): | -0.030 | -0.009 | 0.018 | -0.030 |
| TNS (ns): | -0.439 | -0.040 | 0.000 | -0.400 |
| Violating Paths: | 26 | 9 | 0 | 17 |
| All Paths: | 115 | 39 | 4 | 107 |
+-----+-----+-----+-----+

Density: 61.767%

Reported timing to dir ./timingReports
Total CPU time: 0.81 sec
Total Real time: 1.0 sec
Total Memory Usage: 2979.300781 Mbytes
Reset AAE Options
*** time_design #8 [finish] : cpu/real = 0:00:00.8/0:00:01.4 (0.6), totSession cpu/real = 0:03:44.6/0:18:19.7 (0.2), mem = 2979.3M
0
@Uninuous 76%
cadence-server.vt.ac.in 4% 59.26 GB / 1006.85 GB 0.05 Mb/s 0.01 Mb/s 50 days root 24mvd0003 (x2) mne22497 23bec0058 23bec0325 23bec0059 23bec0444 23bm000
```

iii)optimization

```
cadence.vt.ac.in (23bvd0030)
Setup views included:
wc
Hold views included:
bc

+-----+-----+-----+-----+
| Setup mode | all | reg2reg | reg2cgate | default |
+-----+-----+-----+-----+
| WNS (ns): | -0.006 | 0.008 | 0.041 | -0.006 |
| TNS (ns): | -0.022 | 0.000 | 0.000 | -0.022 |
| Violating Paths: | 5 | 0 | 0 | 5 |
| All Paths: | 115 | 39 | 4 | 107 |
+-----+-----+-----+-----+

+-----+-----+-----+-----+
| Hold mode | all | reg2reg | reg2cgate | default |
+-----+-----+-----+-----+
| WNS (ns): | -0.010 | 0.000 | 0.028 | -0.010 |
| TNS (ns): | -0.013 | 0.000 | 0.000 | -0.013 |
| Violating Paths: | 2 | 0 | 0 | 2 |
| All Paths: | 115 | 39 | 4 | 107 |
+-----+-----+-----+-----+

+-----+-----+-----+-----+
| DRV's | Real | Total |
+-----+-----+-----+-----+
| Nr nets (terms) | Worst Vio | Nr nets (terms) |
+-----+-----+-----+-----+
| max_cap | 0 (0) | 0.000 | 0 (0) |
| max_tran | 0 (0) | 0.000 | 0 (0) |
| max_fanout | 0 (0) | 0 | 0 (0) |
| max_length | 0 (0) | 0 | 0 (0) |
+-----+-----+-----+-----+

Density: 62.325%
Routing Overflow: 0.00% H and 0.00% V
***opt_design ... cpu = 0:00:15, real = 0:00:17, mem = 2662.2M, totSessionCpu=0:04:00 ***
*** Finished opt_design ***
cadence-server.vt.ac.in 3% 59.44 GB / 1006.85 GB 2.60 Mb/s 0.60 Mb/s 50 days root 24mvd0003 (x2) mne22497 23bec0058 23bec0325 23bec0059 23bec0444 23bm000
```

iv)time design after optimization

a)setup

time_design -post_route -setup

```
cadence.vti.ac.in (23bvd0030)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split Multitex Tunneling Packages Settings Help
Quick connect...
home\userdata\23bvd0030\asic.v
Name
Remote monitoring
Follow terminal folder
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net

time_design Summary

Setup views included:
wc

+-----+-----+-----+-----+
| Setup mode | all | reg2reg | reg2cgate | default |
+-----+-----+-----+-----+
| WNS (ns):  | -0.044 | 0.052 | 0.147 | -0.044 |
| TNS (ns):  | -0.207 | 0.000 | 0.000 | -0.207 |
| Violating Paths: | 9 | 0 | 0 | 9 |
| All Paths: | 115 | 39 | 4 | 107 |
+-----+-----+-----+-----+

+-----+-----+-----+-----+
| DRV | Real | Total |
|-----+-----+-----+-----+
| Nr nets(terms) | Worst Vio | Nr nets(terms) |
+-----+-----+-----+-----+
| max_cap | 0 (0) | 0.000 | 0 (0) |
| max_tran | 0 (0) | 0.000 | 0 (0) |
| max_fanout | 0 (0) | 0 | 0 (0) |
| max_length | 0 (0) | 0 | 0 (0) |
+-----+-----+-----+-----+

Density: 62.325%

Reported timing to dir ./timingReports
Total CPU time: 5.04 sec
Total Real time: 14.0 sec
Total Memory Usage: 3003.492188 Mbytes
*** time_design #9 [finish] : cpu/real = 0:00:05.1/0:00:13.5 (0.4), totSession cpu/real = 0:04:07.9/0:20:44.1 (0.2), mem = 3003.5M
@unnovus 78>
```

b)hold

time_design -post_route -hold

```
cadence.vti.ac.in (23bvd0030)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split Multitex Tunneling Packages Settings Help
Quick connect...
home\userdata\23bvd0030\asic.v
Name
Remote monitoring
Follow terminal folder
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net

Loading CTE timing window with TwFlowType 0...(CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 3073.8M)
Add other clocks and setupCtetoAAEClockMapping during iter 1
Loading CTE timing window is completed (CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 3073.8M)
Starting SI iteration 2
Start delay calculation (fullDC) (1 T). (MEM=3012.87)
Glitch Analysis: View bc -- Total Number of Nets Skipped = 4.
Glitch Analysis: View bc -- Total Number of Nets Analyzed = 482.
Total number of fetched objects 482
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 485, 36.3 percent of the nets selected for SI analysis
End delay calculation. (MEM=3055.55 CPU=0:00:00.0 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=3055.55 CPU=0:00:00.0 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:00.5 real=0:00:01.0 totSessionCpu=0:04:09 mem=3079.6M)

time_design Summary

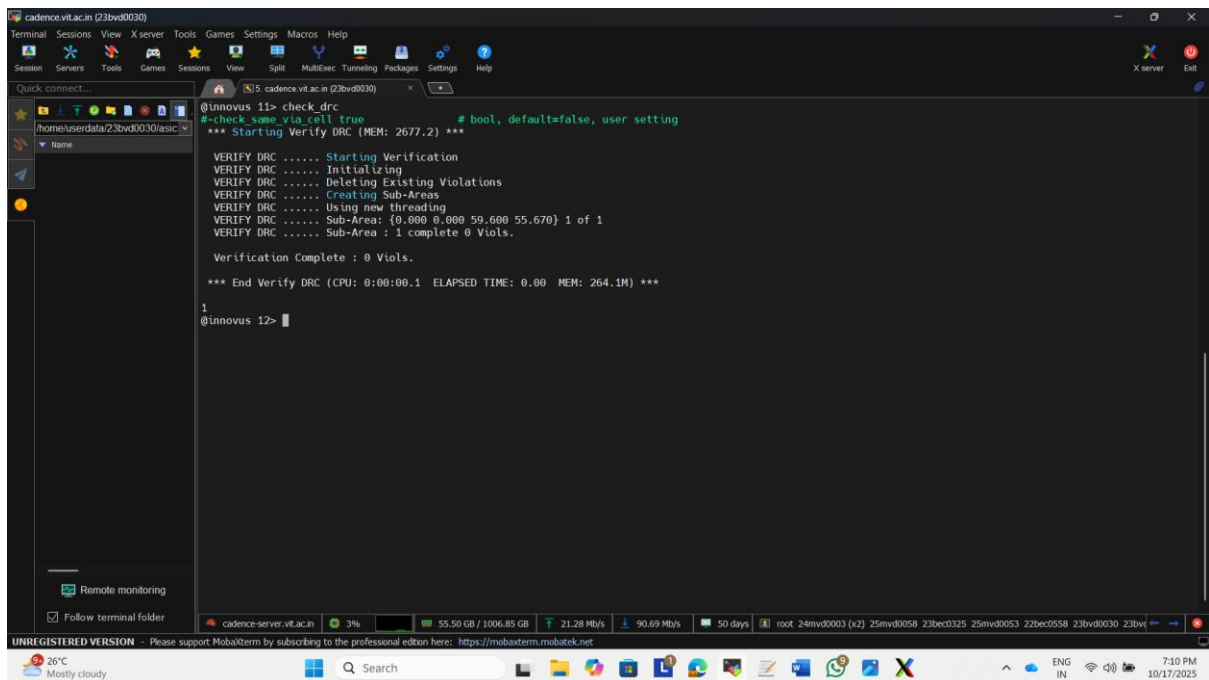
Hold views included:
bc

+-----+-----+-----+-----+
| Hold mode | all | reg2reg | reg2cgate | default |
+-----+-----+-----+-----+
| WNS (ns):  | -0.030 | -0.010 | 0.017 | -0.030 |
| TNS (ns):  | -0.443 | -0.040 | 0.000 | -0.402 |
| Violating Paths: | 25 | 8 | 0 | 17 |
| All Paths: | 115 | 39 | 4 | 107 |
+-----+-----+-----+-----+

Density: 62.325%

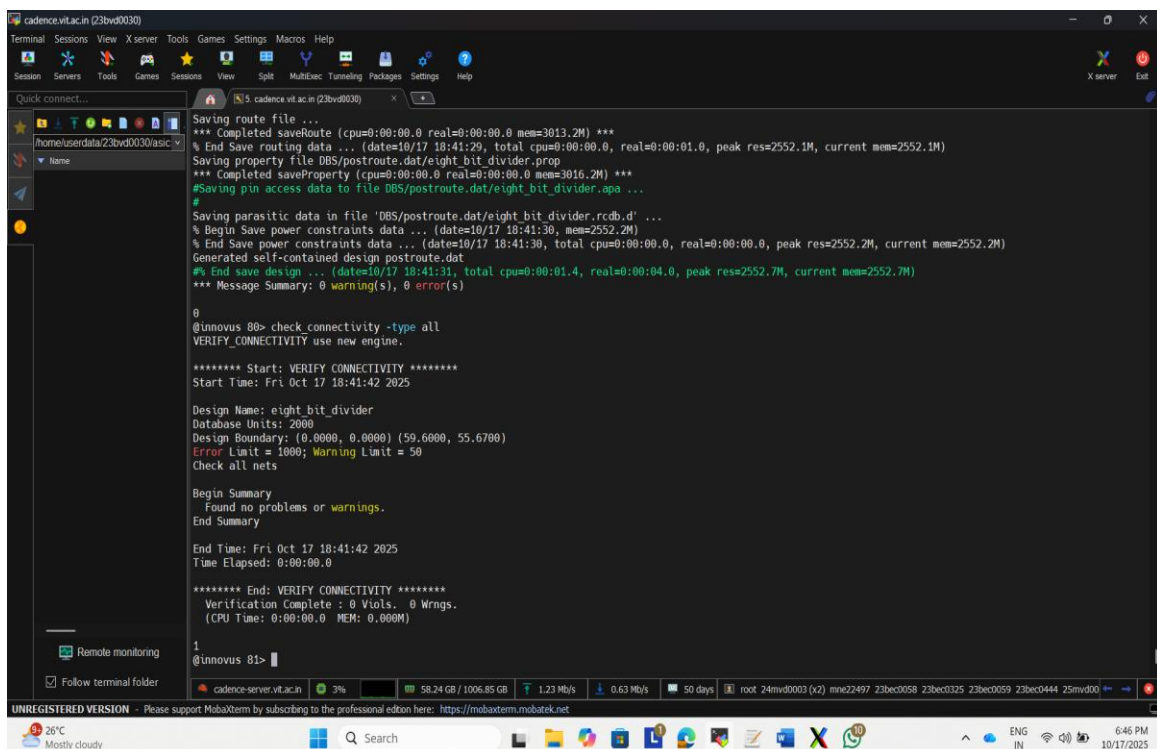
Reported timing to dir ./timingReports
Total CPU time: 0.82 sec
Total Real time: 2.0 sec
Total Memory Usage: 3012.582031 Mbytes
Reset AAE Options
*** time_design #10 [finish] : cpu/real = 0:00:00.8/0:00:01.4 (0.6), totSession cpu/real = 0:04:09.3/0:21:09.2 (0.2), mem = 3012.6M
@unnovus 79>
```

Check_drc



```
cadence.vt.ac.in (23bvd0030)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
home/userdata/23bvd0030/asic
Name
Remote monitoring
Follow terminal folder
@innovus 11> check_drc
#-check_same_via_cell true
*** Starting Verify DRC (MEM: 2677.2) ***
Verify DRC ..... Starting Verification
Verify DRC ..... Initializing
Verify DRC ..... Deleting Existing Violations
Verify DRC ..... Creating Sub-Areas
Verify DRC ..... Using new threading
Verify DRC ..... Sub-Area: {0.000 0.000 59.600 55.670} 1 of 1
Verify DRC ..... Sub-Area : 1 complete 0 Viols.
Verification Complete : 0 Viols.
*** End Verify DRC (CPU: 0:00:00.1 ELAPSED TIME: 0.00 MEM: 264.1M) ***
1
@innovus 12>
```

Check_connectivity



```
cadence.vt.ac.in (23bvd0030)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
home/userdata/23bvd0030/asic
Name
Remote monitoring
Follow terminal folder
Saving route file ...
*** Completed saveRoute (cpu=0:00:00.0 real=0:00:00.0 mem=3013.2M) ***
% End Save routing data ... (date=10/17 18:41:29, total cpu=0:00:00.0, real=0:00:01.0, peak res=2552.1M, current mem=2552.1M)
Saving property file DBS/postroute.dat/eight_bit_divider.prop
*** Completed saveProperty (cpu=0:00:00.0 real=0:00:00.0 mem=3016.2M) ***
#Saving pin access data to file DBS/postroute.dat/eight_bit_divider.apa ...
#
Saving parasitic data in file 'DBS/postroute.dat/eight_bit_divider.rcdb.d' ...
% Begin Save power constraints data ... (date=10/17 18:41:30, mem=2552.2M)
% End Save power constraints data ... (date=10/17 18:41:30, total cpu=0:00:00.0, real=0:00:00.0, peak res=2552.2M, current mem=2552.2M)
Generated self-contained design postroute.dat
#% End save design ... (date=10/17 18:41:31, total cpu=0:00:01.4, real=0:00:04.0, peak res=2552.7M, current mem=2552.7M)
*** Message Summary: 0 warning(s), 0 error(s)
0
@innovus 80> check_connectivity -type all
VERIFY_CONNECTIVITY use new engine.
***** Start: VERIFY CONNECTIVITY *****
Start Time: Fri Oct 17 18:41:42 2025
Design Name: eight_bit_divider
Database Units: 2080
Design Boundary: (0.0000, 0.0000) (59.6000, 55.6700)
Error Limit = 1000; Warning Limit = 50
Check all nets
Begin Summary
Found no problems or warnings.
End Summary
End Time: Fri Oct 17 18:41:42 2025
Time Elapsed: 0:00:00.0
***** End: VERIFY CONNECTIVITY *****
Verification Complete : 0 Viols. 0 Wrngs.
(CPU Time: 0:00:00.0 MEM: 0.000M)
1
@innovus 81>
```

Post_route

